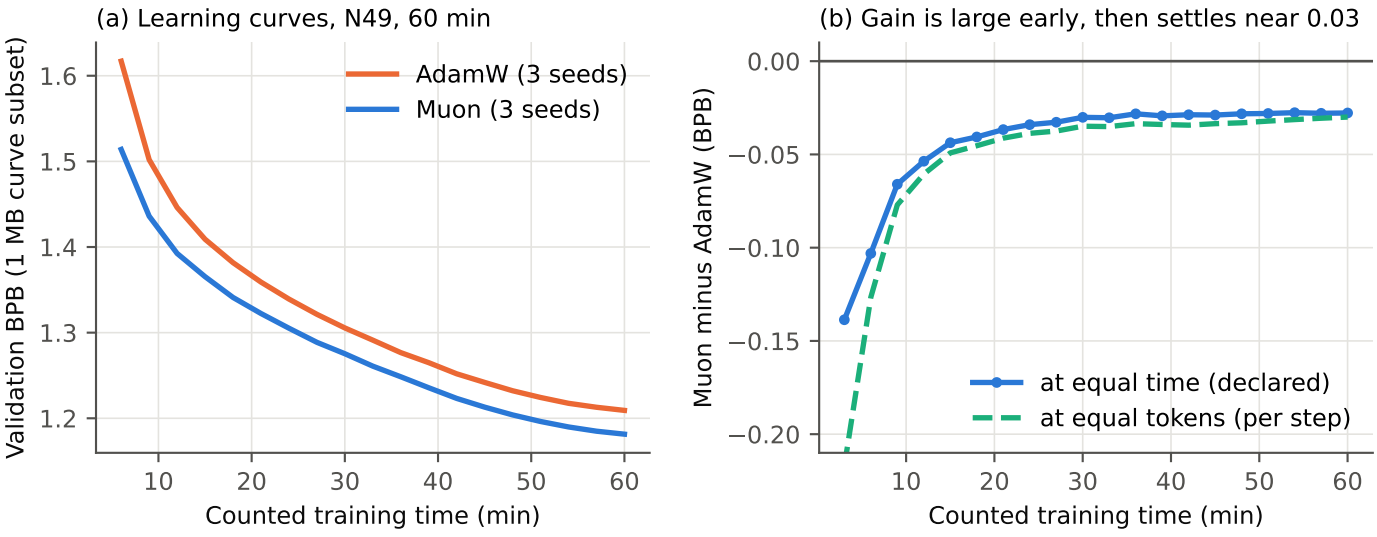
(a) Learning curves, N49, 60 min
AdamW (3 seeds)
Muon (3 seeds)
Validation BPB (1 MB curve subset)
1.6
1.5
1.4
1.3
1.2
10
20
30
40
50
60
Counted training time (min)
(b) Gain is large early, then settles near 0.03
Muon minus AdamW (BPB)
0.00
−0.05
−0.10
−0.15
−0.20
at equal time (declared)
at equal tokens (per step)
10
20
30
40
50
60
Counted training time (min)